

Corrine Xiang

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SUMMARY

- **Platform & Backend:** 3 software engineering internships building cloud-native backend systems, platform tooling, and agentic AI automations, with hands-on experience in Python, Java, AWS, APIs, and distributed data systems
- **Business-minded Engineering:** Computer Science major with a Rotman School of Management background, strong at interpreting requirements, clarifying ambiguities, and evaluating technical trade-offs

EDUCATION

University of Toronto

Toronto, ON

Bachelor of Science, Computer Science Major + Business Minor

Exp. Jun 2028

- GPA: 3.6/4.0, Dean's List 2022-2025

SKILLS

Languages & OS: Python, Java, C, SQL, HTML/CSS, JavaScript/TypeScript, Bash, Linux

Backend & Frontend: Spring Boot, Node.js, Next.js, React, Flask, Django, REST / OpenAPI, JUnit, Maven

Database & Big Data: Redis, PostgreSQL, MongoDB, Apache Spark, Apache Flink, Apache Iceberg

Cloud & DevOps: Git, AWS (IAM, S3, SQS, EMR, Athena, DynamoDB), GCP (Cloud Storage, Cloud Run), Docker

EXPERIENCE

Software Engineering Intern - Platform

Jan 2026 – Aug 2026

Terminal

Toronto, ON

- Developed an event-driven data validation feature for new-provider onboarding, publishing from **Lambda** through **SQS** to model-specific **Spring Boot** handlers, supporting **290+** providers and **19+** canonical data models
- Designed a 3-tier deduplication strategy: **Redis** hot cache, **AWS Athena/Iceberg** cold lookup with cache write-back, and Temporal workflow-ID idempotency, to prevent redundant validation runs
- Spiked automating scheduled investigations with **Cursor Automations**, assessing prompt/skill effectiveness, orchestration approaches (agent framework vs single-skill), and query-layer trade-offs (Athena MCP vs Spark) — producing a technical findings doc adopted by the team
- Provisioned scoped **IAM** roles and **S3** bucket policies (Athena + AWS Glue + Iceberg access), enabling Cursor Background Agent to run hourly scheduled investigations in a testing environment
- Triaged on-call alerts and resolved incidents, including a 2-year data gap caused by JavaBean naming bug in Flink CDC deserialization, and wrote a PySpark backfill to repair historical records

Software Engineering Intern - AI/IoT

May 2025 – Aug 2025

BSH Home Appliances Corporation

Remote

- Tuned the input-validation filter in an **AI/IoT Python** system for **refrigerator R&D testing**, increasing downstream food freshness detection accuracy from **73% to 77%** by reducing false rejections of valid food items
- Extended the R&D project's data delivery pipeline with new **AWS S3** and **HTTP API** channels — drawer-close events trigger multi-sensor capture, on-device YOLOX inference, and multi-channel delivery — achieving **under 1.3-second end-to-end latency** from scan to S3 upload
- Improved system reliability with **shell scripts**, **cron scheduling**, and structured logging, including fixing a date-rollover bug and migrating restart control from hardcoded daily device reboots to **systemd**-managed services
- Prototyped a multi-interface monitoring system to detect refrigerator test anomalies and send SMTP email alerts, implementing Flask web interface, Tkinter GUI, and CLI tools, reducing engineer troubleshooting time **by 20%**
- Designed monitor system's Python backend services with **Clean Architecture**, including **TCP**-based file shipment, a configurable YAML-based rule engine for anomaly detection, and real-time alerting/logging modules

Software Development Intern - Web

Sep 2024 – Dec 2024

Re:Pair Genomics Inc.

Toronto, ON

- Built a full-stack Human Resources Management System with Next.js 15, React 18, and TypeScript, implementing user authentication, role-based access control, and timesheet reporting via Redux Toolkit
- Implemented a cloud-native serverless architecture with **AWS Lambda**, integrated with **DynamoDB** for data storage and SST for cloud deployment, reducing the projected monthly operating cost by **\$110** per employee

PROJECTS

Return AI | Python, Flask, SQLAlchemy, PostgreSQL, React, Docker, RESTful API, Figma Feb 2025 – Mar 2025

- Developed an AI-powered return processing system using Google **Gemini API** and **NLP** tools, analyzing images and descriptions with **88%+** categorization accuracy and reliable recommendations (**90% confidence level**)
- Implemented a **RESTful API** with JWT authentication and OpenAPI documentation, enabling server-to-server communication between e-commerce platforms and the system, reducing the return handling time by 30%